Name
Date
Course

Practice Beginning Algebra Test III (Version 2)

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.

Solve.

1) $\begin{cases} -6x+y=2 \\ 6x-3y=0 \end{cases}$

2) $\begin{cases} y-x=2 \\ x=1-y \end{cases}$

3) $\begin{cases} -2x+3=y \\ 3x+4y=-5 \end{cases}$

4) $\begin{cases} y=-7x \\ 4y+2x=1 \end{cases}$

5) Identify the form of each linear equation below by writing "S" for standard, "SI" for slope-intercept, or "PS" for point-slope.

a) $y=x-\frac{1}{2}$

b) $8=3x+10y$

c) $\frac{1}{4}x+0.0007=y$

d) $y-2=\frac{4}{3}(x+111)$

6) a) Use the slope equation to find the slope of the line containing the two points below.
b) Then write a point-slope form of the equation of the line containing the two points below.
c) Then convert the equation from point-slope form to slope-intercept form.
d) Find the y-intercept of the line.

$(6,2)$ & $(-4,-18)$

⑦ Use the slope-intercept equation to find the slope-intercept form of the line with the given characteristics.

a) Y-intercept $= 8$, contains the point $(-3,4)$.

b) Slope $= -3$, contains the point $(-5,-3)$.

⑧ For each problem below, write an equation of the line with the given characteristics.

a) Parallel to $y-7=2(x-4)$ and containing the point $(5,2)$.

b) Perpendicular to $\frac{4}{5}x+\frac{2}{7}=y$ and containing the point $(-4,-4)$.

c) Perpendicular to $y+2=-(x+1)$ and containing the point $(1,-1)$.

d) Parallel to $-3=7x-10y$ and containing the point $(-8,-7)$.

⑨ For each problem below, write an equation of the line with the given characteristics. There is more than one way to write the answers, but try to use the method that is quickest.

a) Slope: -7
Y-intercept: 2

b) X-intercept: 9
Slope: $-\frac{1}{2}$

c) X-intercept: -2
Y-intercept: -10

d) Slope: 3
Point: $(1,\frac{5}{7})$

(10) Matthew and Nathan enjoy Strawberry Flakes Cereal for breakfast. This month they ate a total of 18 bowls. Nathan ate three less bowls than twice the number of bowls that Matthew ate. How many bowls did Nathan eat?

(11) Jerry bought four Delicious bars and one Crazy Sour gum ball for $4.47. The next day he bought one Delicious bar and three Crazy Sour gum balls for $2.19. What is the cost per gum ball and per bar?

(12) Yellow stamps cost $0.31 each and green stamps cost $0.62 each. If Lydia bought 100 stamps for $48.98, how many stamps of each color did she buy?

(13) A grasshopper jumps into the air. Its height in feet t seconds after jumping can be modeled by the following equation (if we neglect air resistance).

$$h = -16t^2 + 12t$$

a) Find the height of the grasshopper $\frac{1}{3}$ of a second after jumping.

b) How many seconds after jumping will the grasshopper reach a height of two feet?

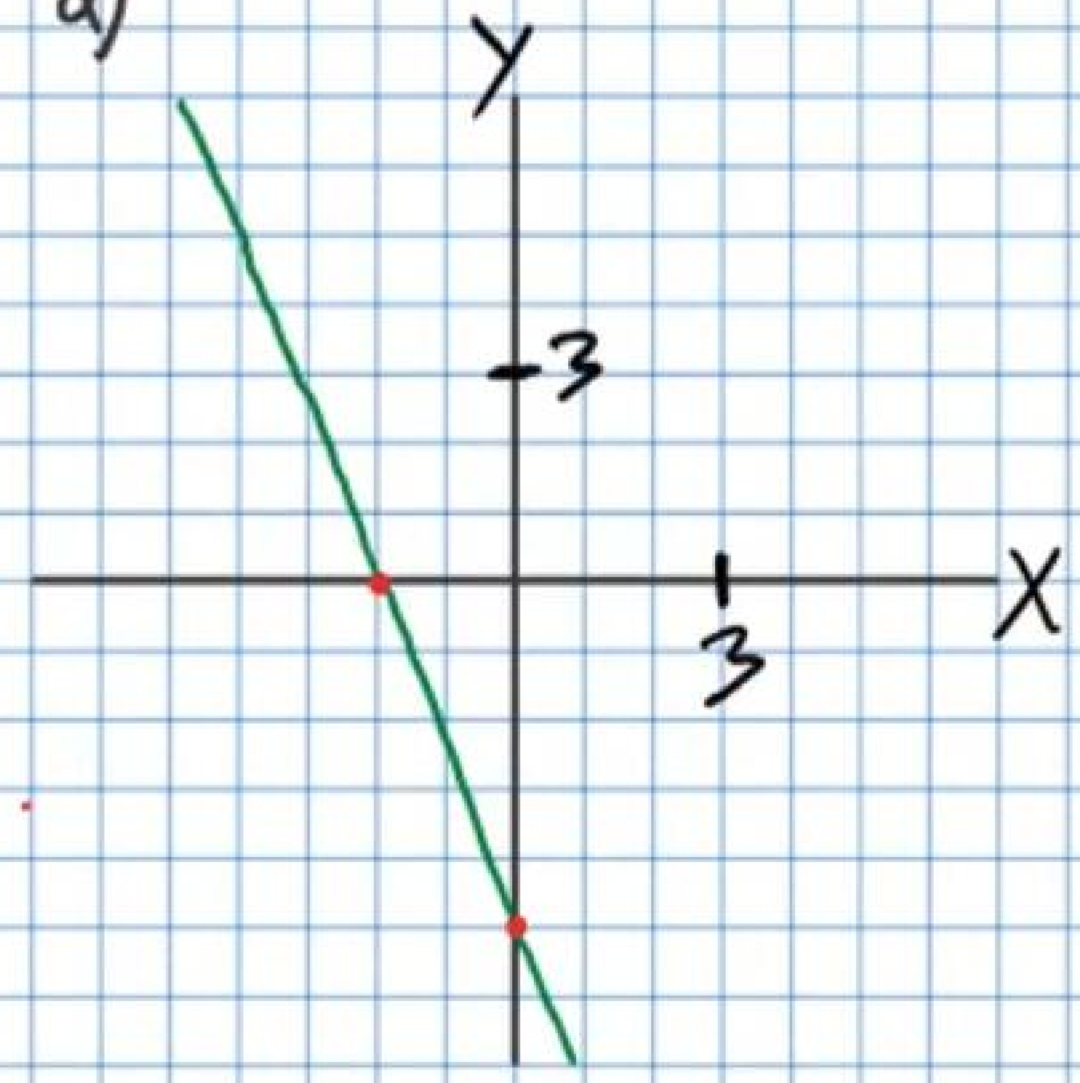
c) How high will the grasshopper be $\frac{3}{8}$ of a second after it jumps?

d) How many seconds after jumping will the grasshopper return to the ground?

For each graph below, write an equation of the line shown.

14

a)



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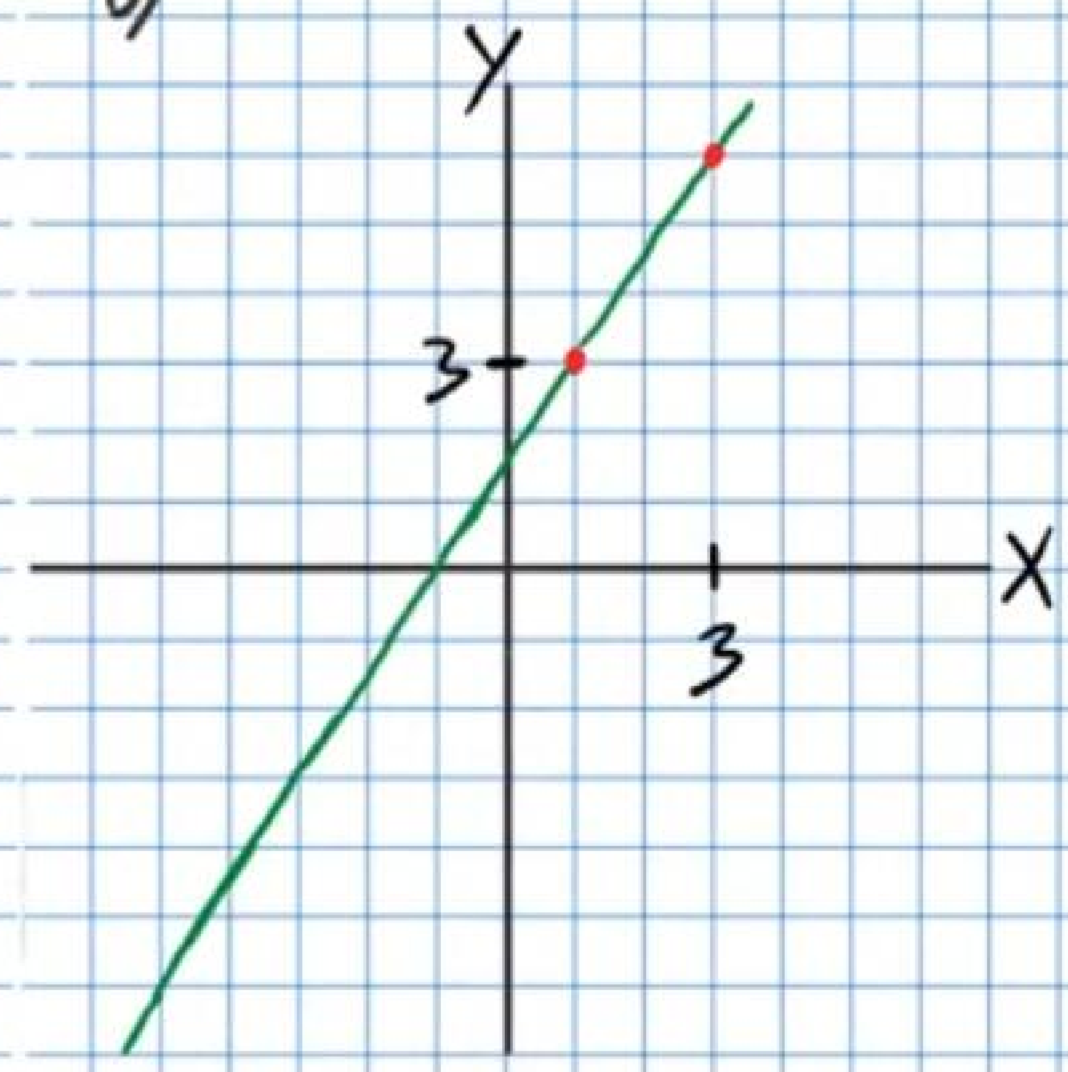

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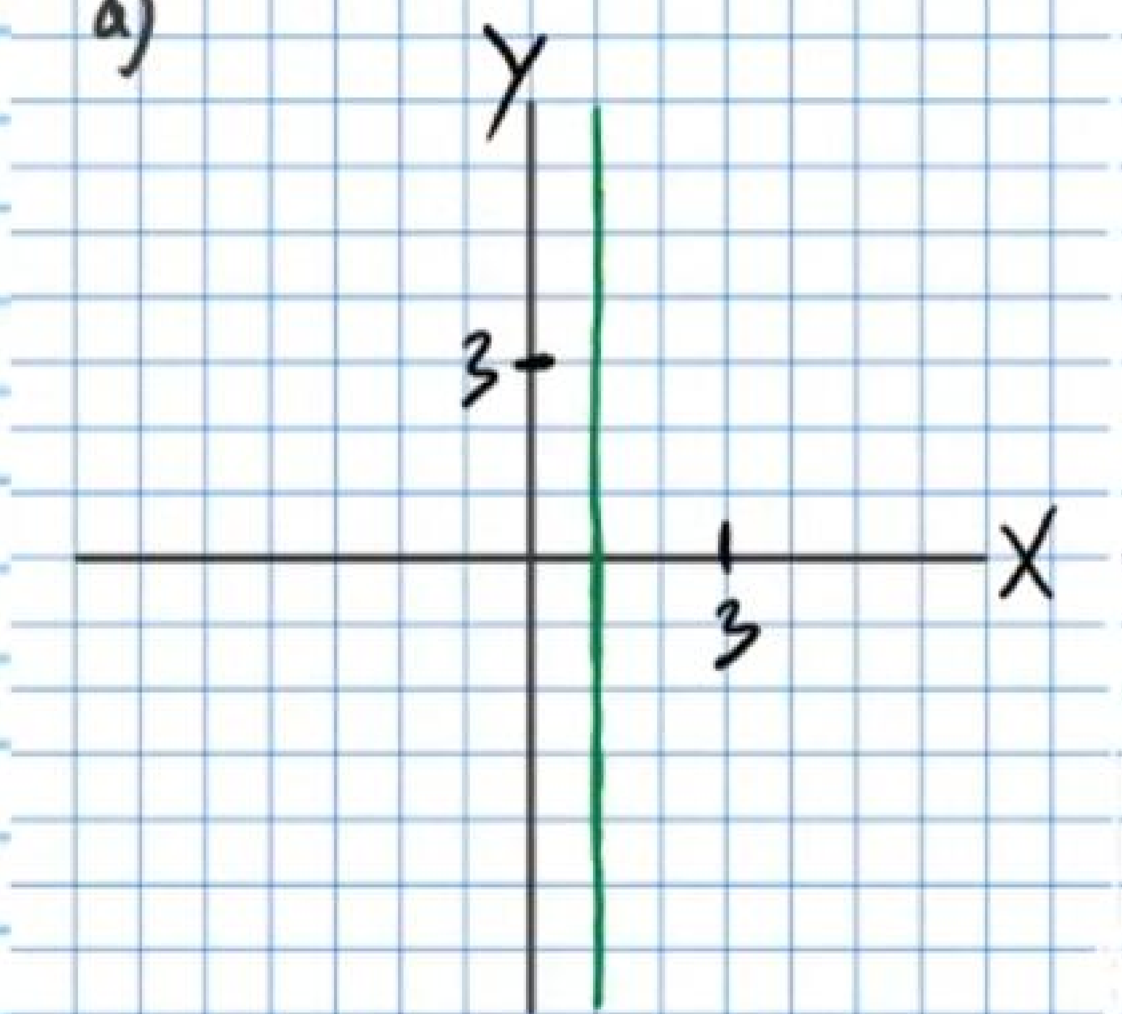
a)



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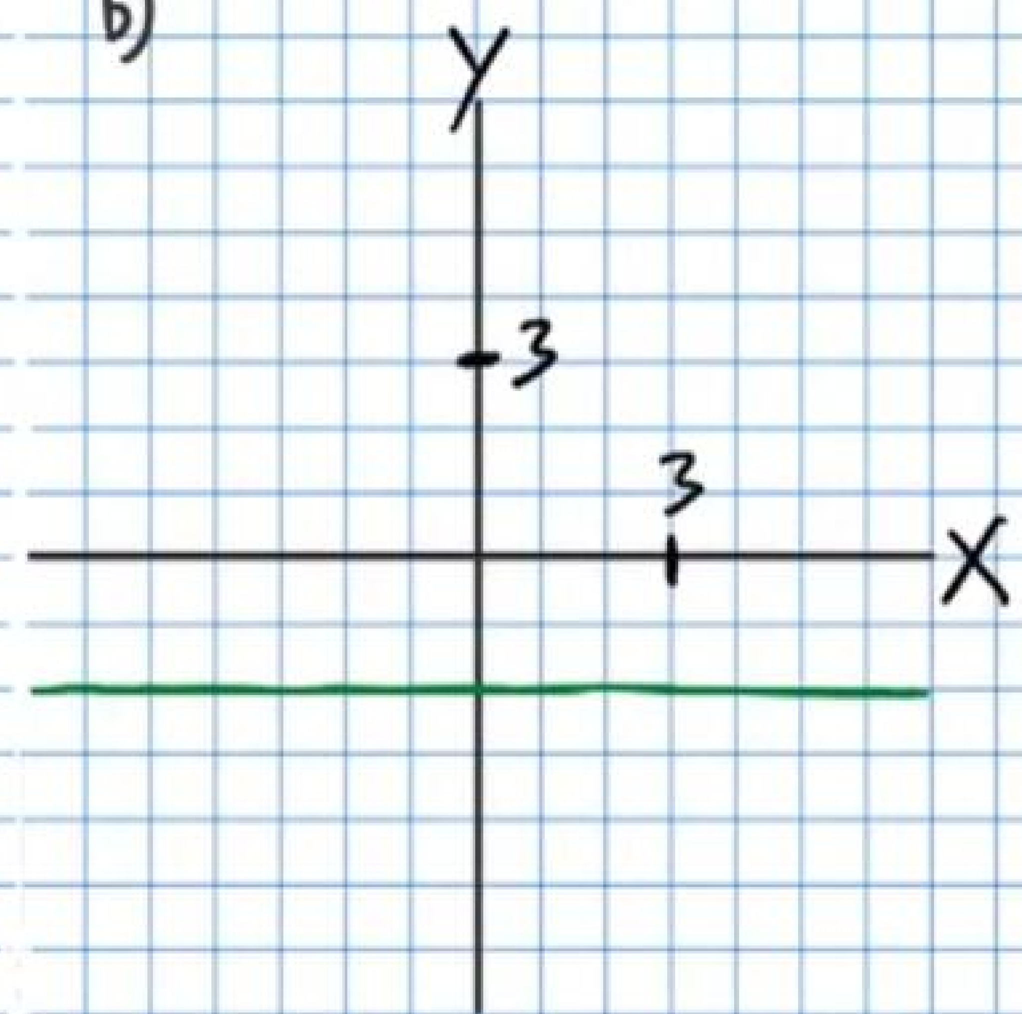


* You must use graphing paper for problem # 16.

(16) a) Solve the system of equations below using the graphing method. As usual, label axes, intervals, and any points you use to solve the system. You might not be able to solve the system using this method, so move on to part b to help you find the answer for part a.

$$\begin{cases} 2x+5y=4 \\ 7x+2y=12 \end{cases}$$

b) Then check your solution by solving the system algebraically (i.e. with substitution or elimination).

For problem #'s 17-20 below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,-3)).

(17) a) Graph the following equation by plotting two points and connecting the dots (Method I).

$$x-2y=4$$

b) Graph the following equation by using x and y intercepts (Method II).

$$-3x+2y=9$$

⑱ Graph the following equations by converting to slope-intercept form (if not already in this form) and using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

a) $3x-2=y$

b) $y=-\frac{x}{2}+1$

⑲ a) Graph the following equation by converting to slope-intercept form (if not already in this form) and using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

$-3x-4y=12$

b) Graph the equation below by using its slope and a point on the line that is visible in the equation (Method IV).

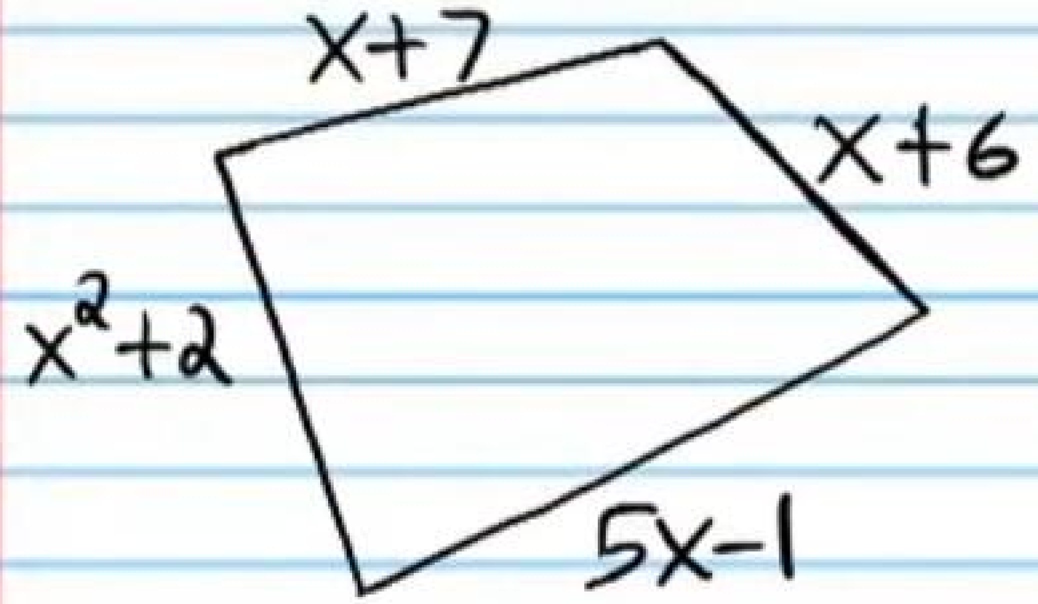
$\frac{2}{3}(x-4)=y+3$

⑳ Graph the following equations.

a) $y=-4$

b) $x=2$

(21) If the perimeter of the quadrilateral below is 44 centimeters, find the length of each side.

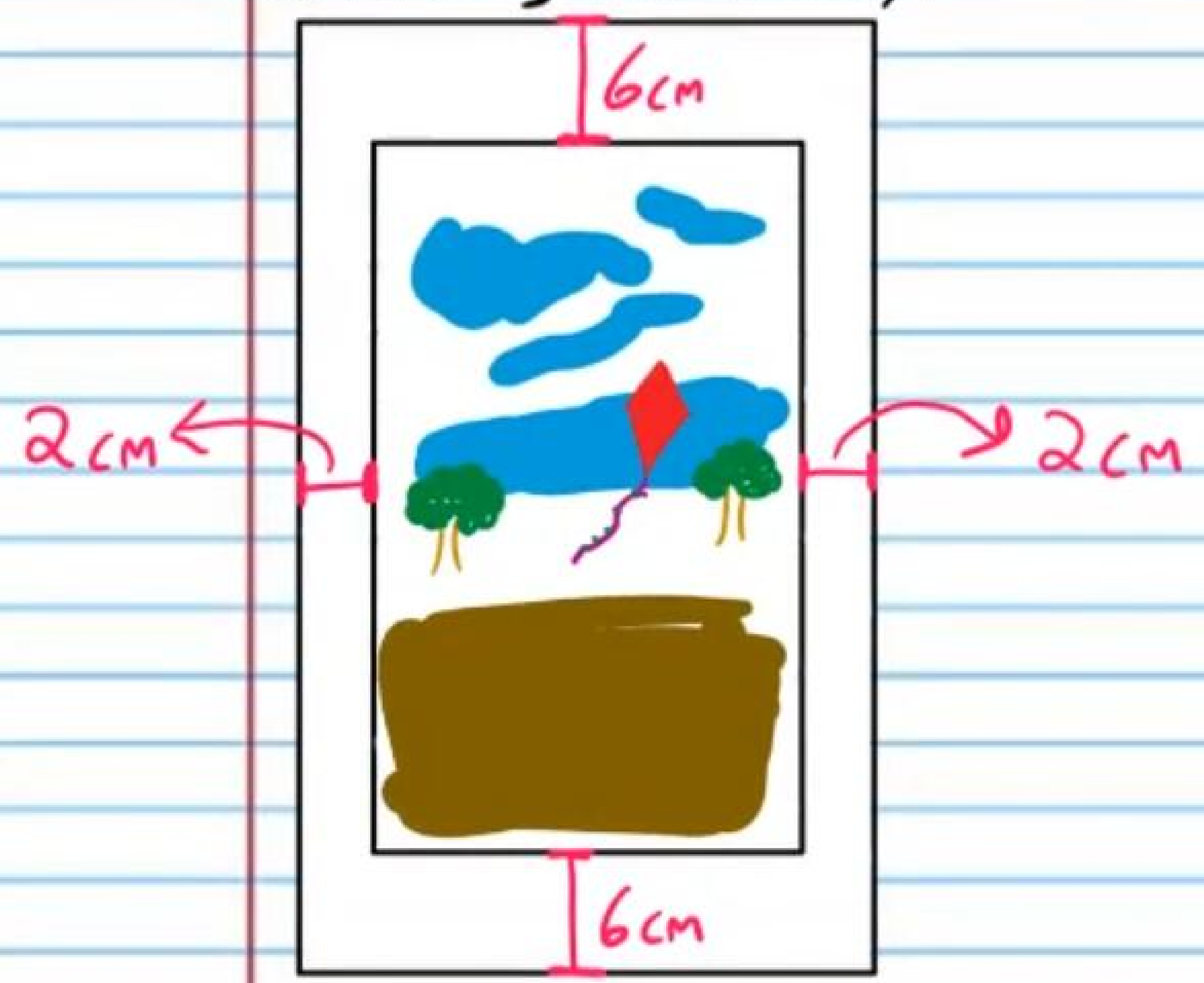


(22) John has quarters and nickels in his pocket. If he has 23 coins in his pocket, and the total worth of these coins is \$3.95, how many of each type of coin does he have?

(23) Linsey wants to make 9 liters of a 72% fruit punch mixture. If she has a 50% fruit punch mixture and a 90% fruit punch mixture, how many liters of each mixture should she combine?

(24) Tommy is going to make 30 ounces of lemonade that costs \$0.23 per ounce. He will mix lemonade that costs \$0.32 per ounce with lemonade that costs \$0.12 per ounce to achieve this. How many ounces of each type of lemonade must he combine?

(25) The area of a rectangular window (including its frame) is 24 more square centimeters than twice the area of the glass part alone. If the width of the rectangular glass part is 9 centimeters shorter than its length, find the dimensions of the entire window (including the frame).



Practice Beginning Algebra Test III (Version 2) Answers

① $\boxed{\left(-\frac{1}{2}, -1\right)}$

② $\boxed{\left(-\frac{1}{2}, \frac{3}{2}\right)}$ or $\boxed{(-0.5, 1.5)}$

③ $\boxed{\left(\frac{17}{5}, -\frac{19}{5}\right)}$ or $\boxed{\left(3\frac{2}{5}, -3\frac{4}{5}\right)}$

④ $\boxed{\left(-\frac{1}{26}, \frac{7}{26}\right)}$

⑤ a) $\boxed{SI}$ b) $\boxed{S}$ c) $\boxed{SI}$ d) $\boxed{PS}$

⑥ a) $\boxed{m=2}$ b) $\boxed{y-2=2(x-6)}$ or $\boxed{y+18=2(x+4)}$
c) $\boxed{y=2x-10}$ d) $\boxed{b=-10}$

⑦ a) $\boxed{y=\frac{4}{3}x+8}$ b) $\boxed{y=-3x-18}$

⑧ a) $\boxed{y-2=2(x-5)}$ or $\boxed{y=2x-8}$

b) $\boxed{y+4=-\frac{5}{4}(x+4)}$ or $\boxed{y=-\frac{5}{4}x-9}$

c) $\boxed{y+1=1(x-1)}$ or $\boxed{y=x-2}$

d) $\boxed{y+7=\frac{7}{10}(x+8)}$ or $\boxed{y=\frac{7}{10}x-\frac{7}{5}}$

9) a) $\boxed{y=-7x+2}$ b) $\boxed{y-0=-\frac{1}{2}(x-9)}$ or $\boxed{y=-\frac{1}{2}x+\frac{9}{2}}$

c) $\boxed{y=-5x-10}$ or $\boxed{y+10=-5(x-0)}$ or $\boxed{y-0=-5(x+2)}$

d) $\boxed{y-\frac{5}{7}=3(x-1)}$ or $\boxed{y=3x-16/7}$

10) $\boxed{\text{11 bowls}}$

11) $\boxed{D=\$1.02/\text{bar}}$ $\boxed{C=\$0.39/\text{ball}}$

12) $\boxed{G=58 \text{ stamps}}$ $\boxed{Y=42 \text{ stamps}}$

13) a) $\boxed{2\frac{2}{9} \text{ ft}}$ b) $\boxed{\frac{1}{4} \text{ seconds } \& \ \frac{1}{2} \text{ seconds}}$

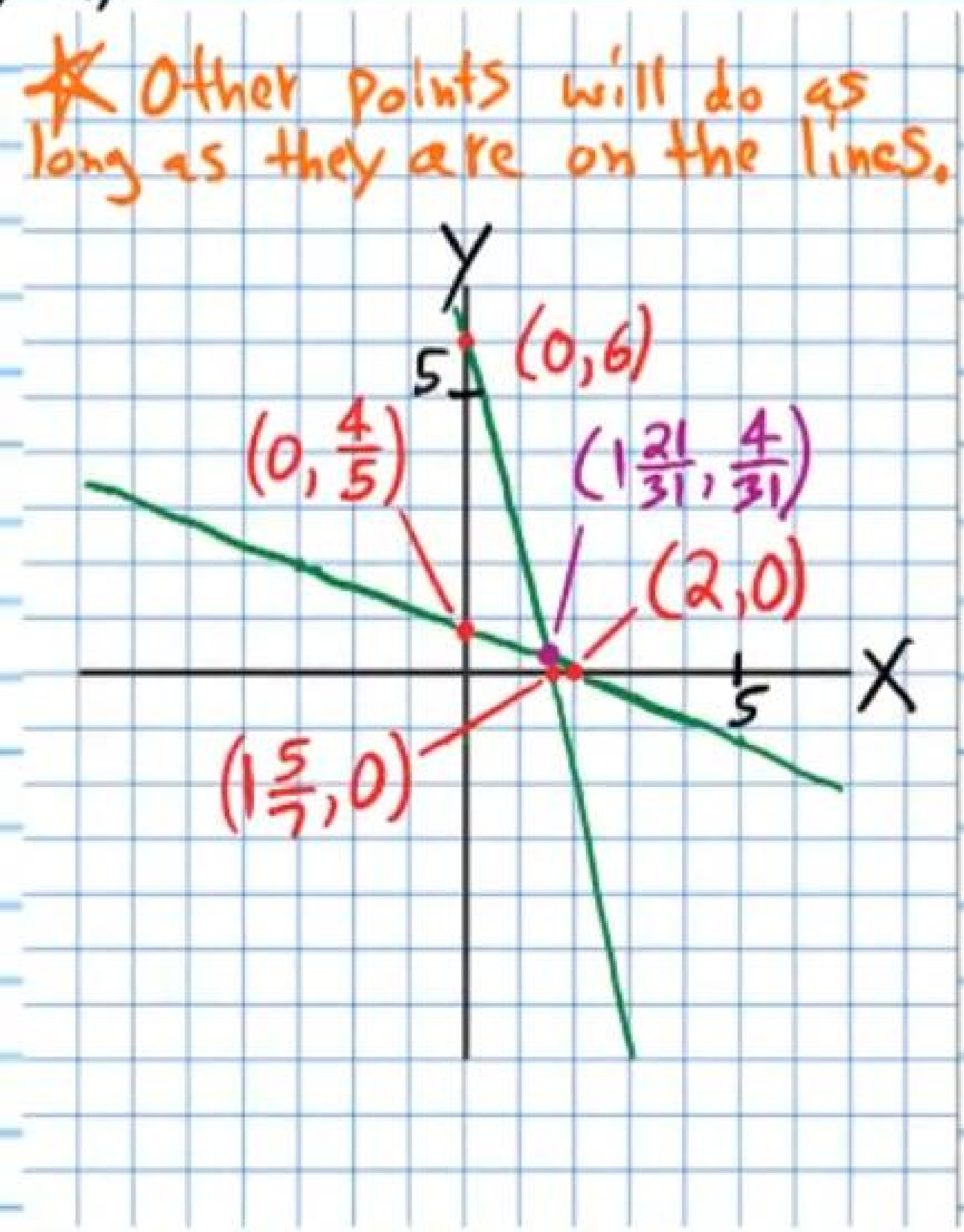
c) $\boxed{\frac{9}{4} \text{ ft}}$ or $\boxed{2.25 \text{ ft}}$ d) $\boxed{3/4 \text{ seconds}}$

14) a) $\boxed{y=-\frac{5}{2}x-5}$ or $\boxed{y+5=-\frac{5}{2}(x-0)}$ or $\boxed{y-0=-\frac{5}{2}(x+2)}$

b) $\boxed{y-3=\frac{3}{2}(x-1)}$ or $\boxed{y-6=\frac{3}{2}(x-3)}$ or $\boxed{y=\frac{3}{2}x+\frac{3}{2}}$

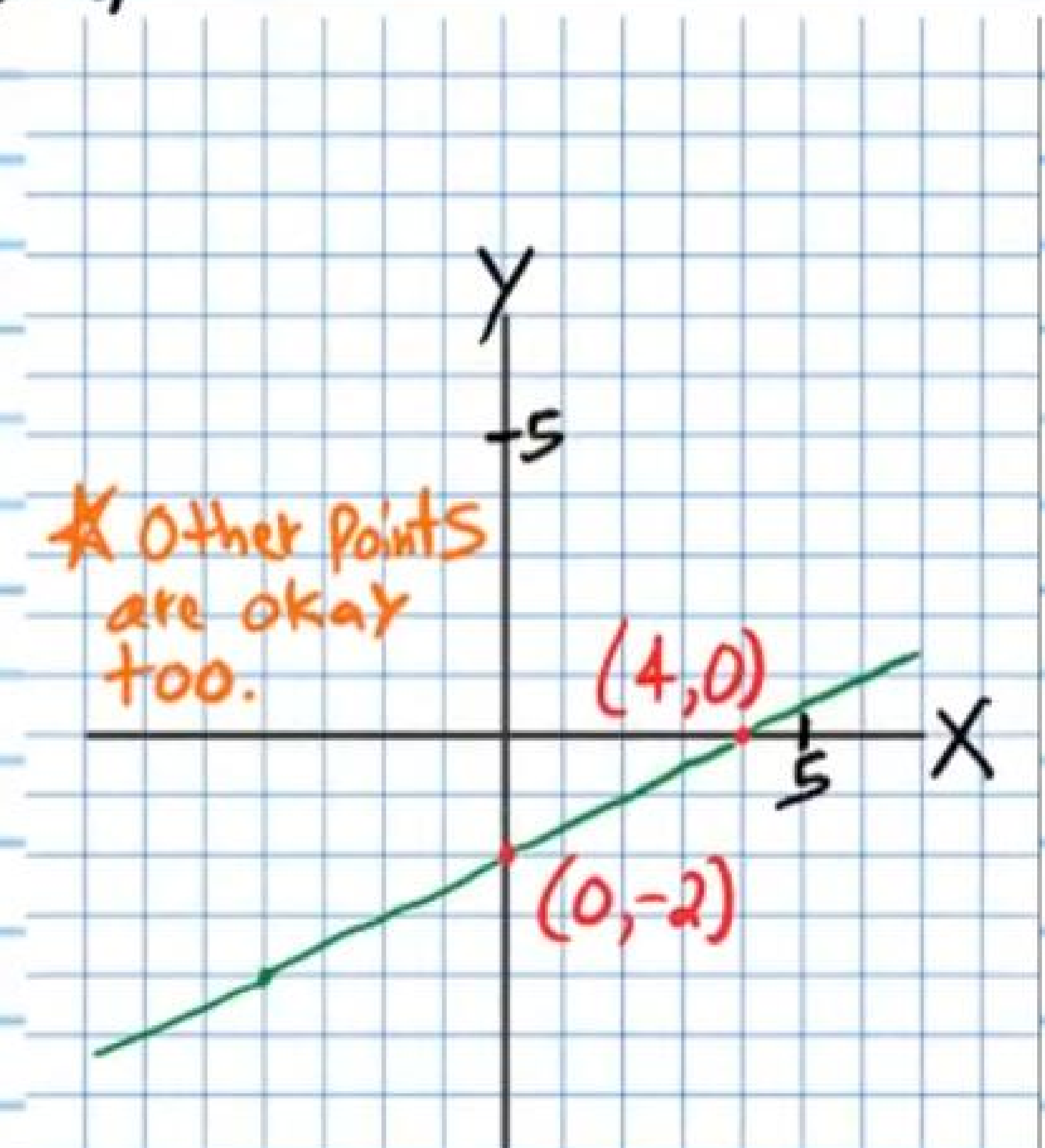
15) a) $\boxed{x=1}$ b) $\boxed{y=-2}$

16) a)



b) $\boxed{x=1\frac{21}{31}}$ $\boxed{y=\frac{4}{31}}$

17) a)



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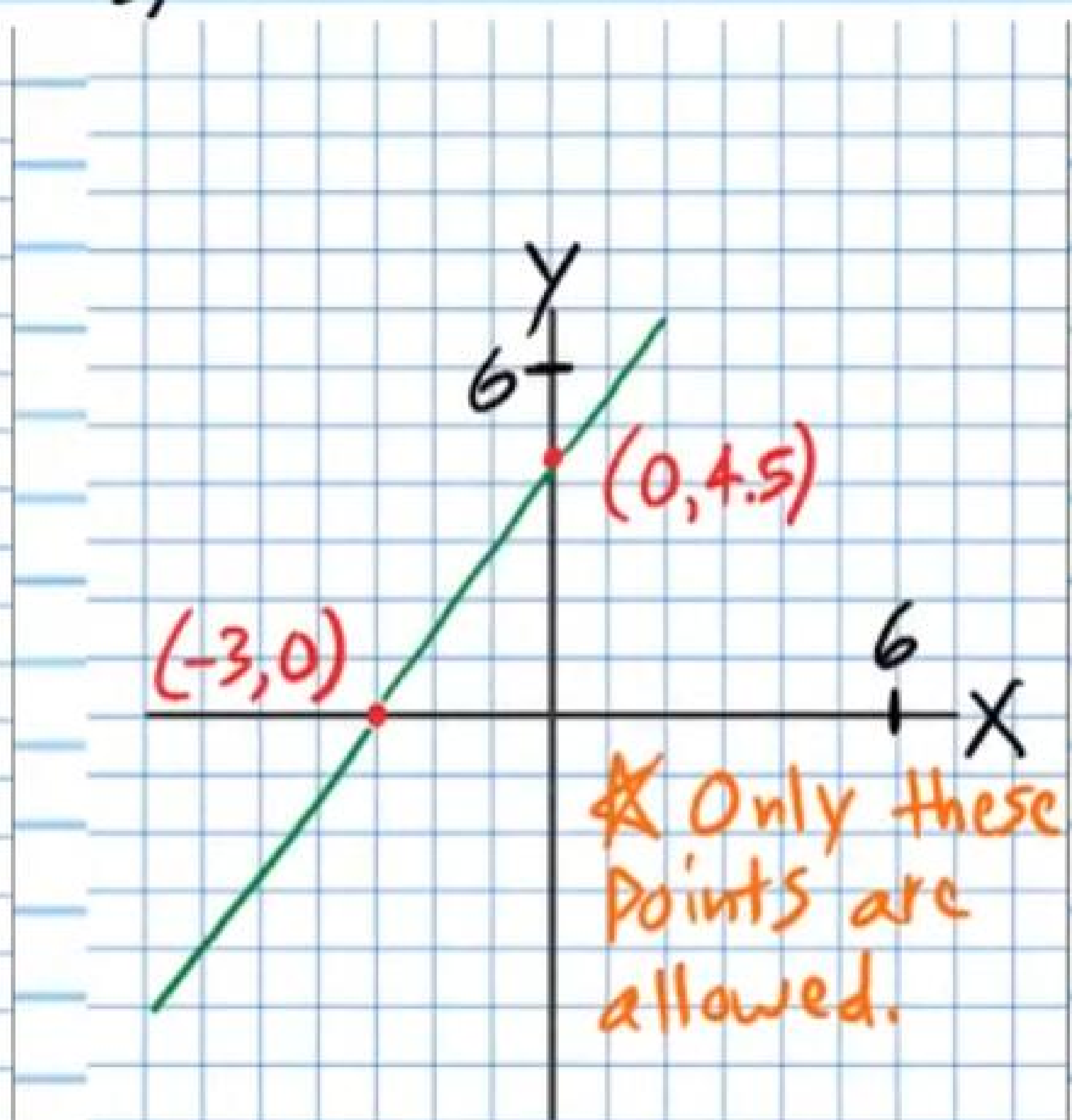

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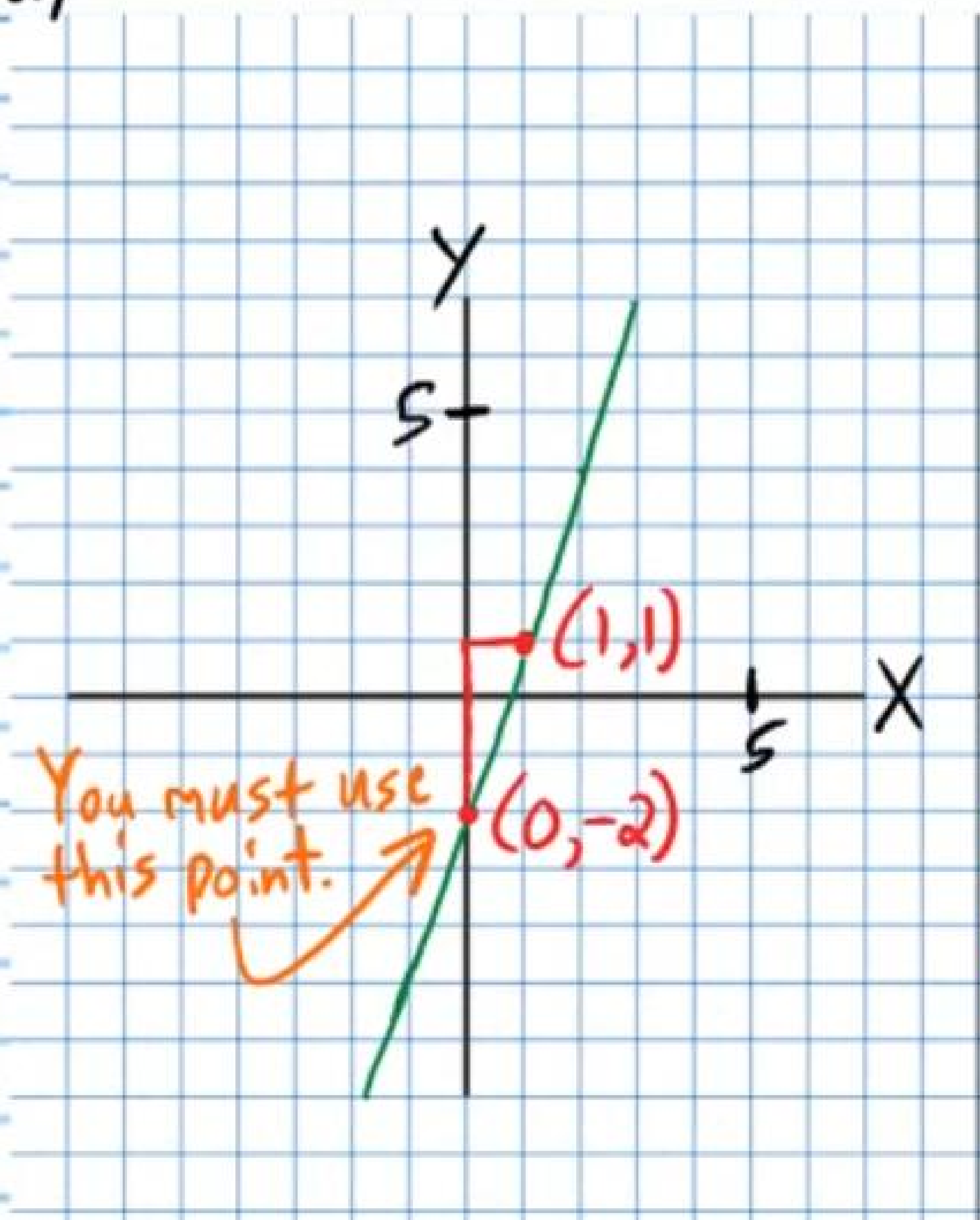

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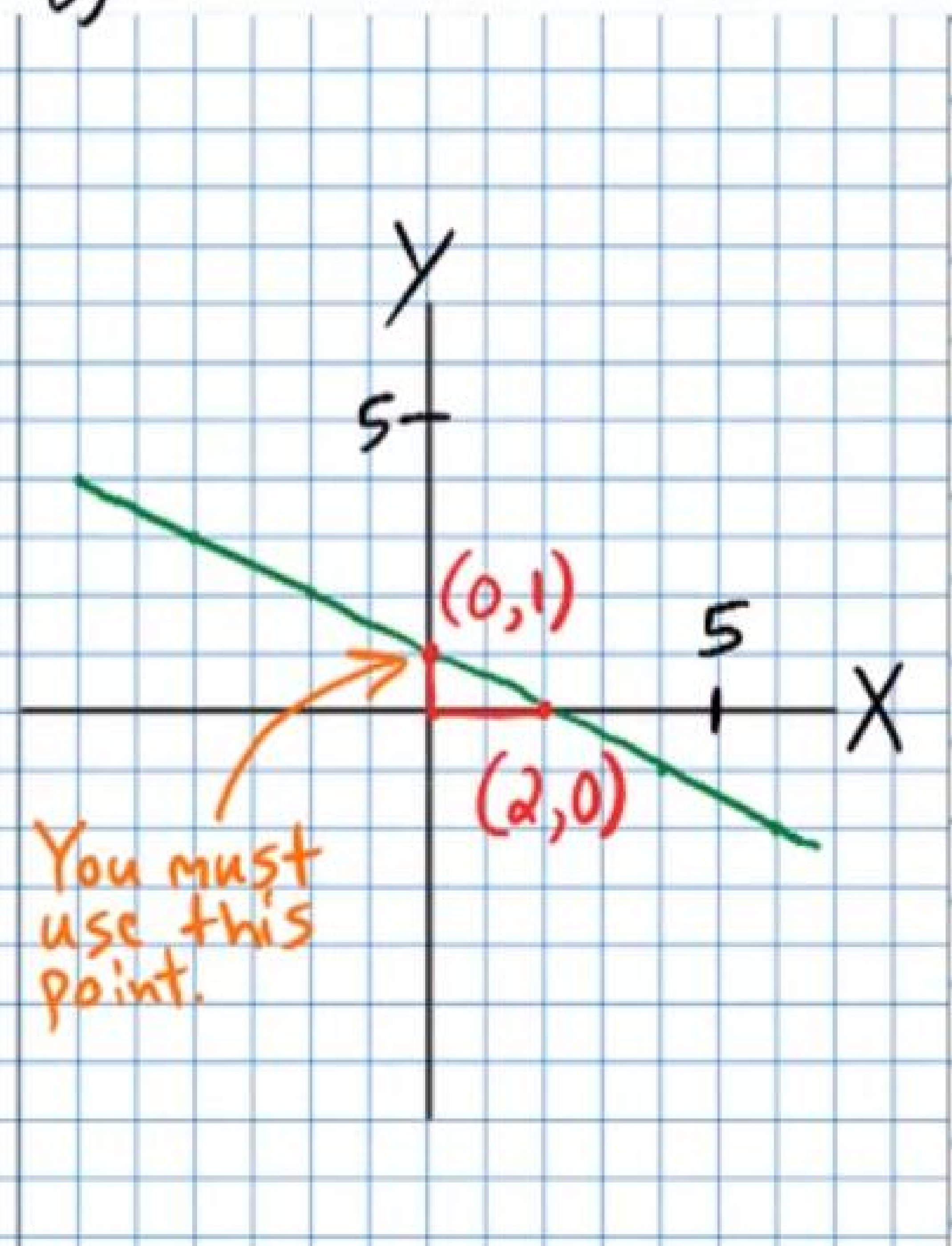
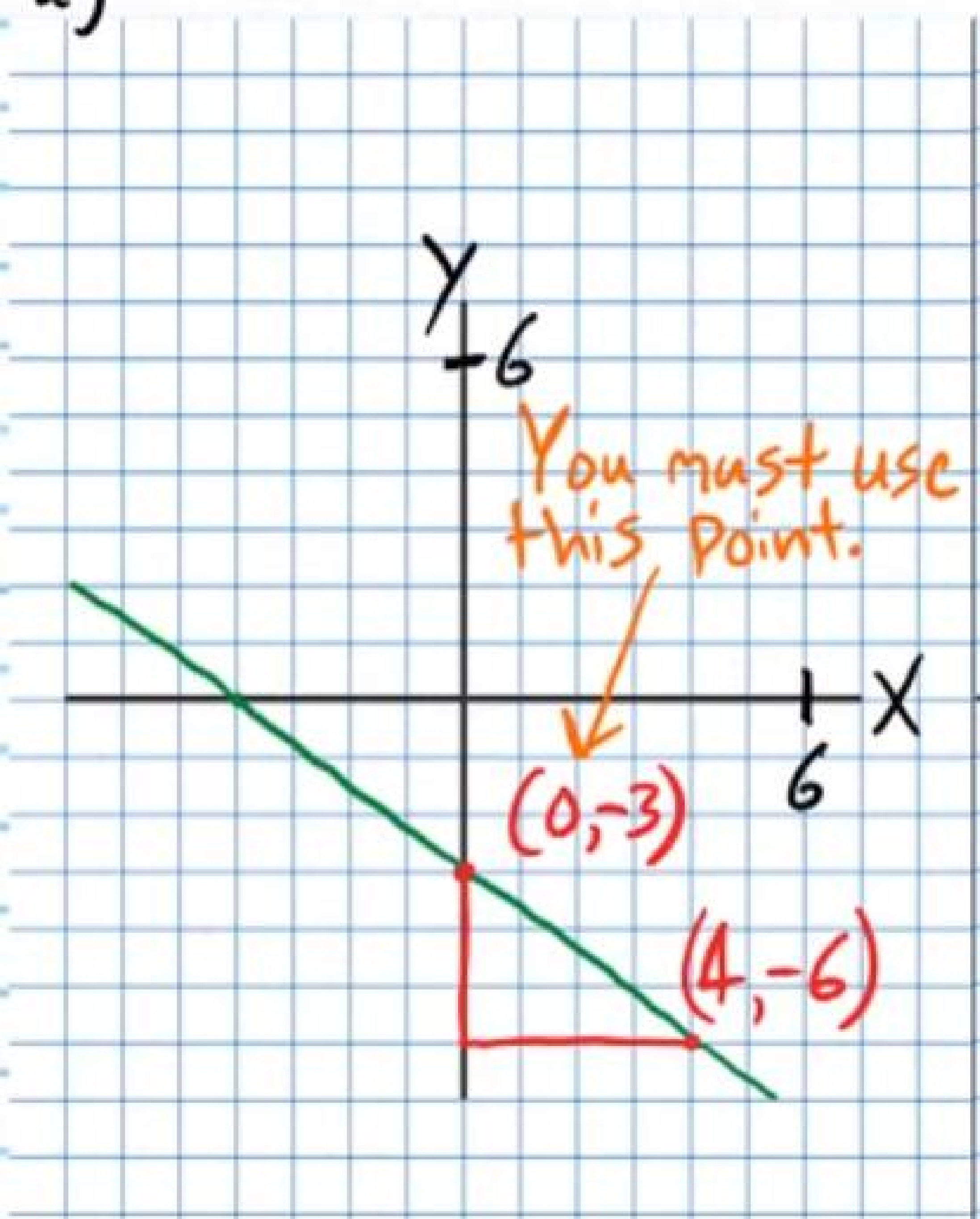


19) a)



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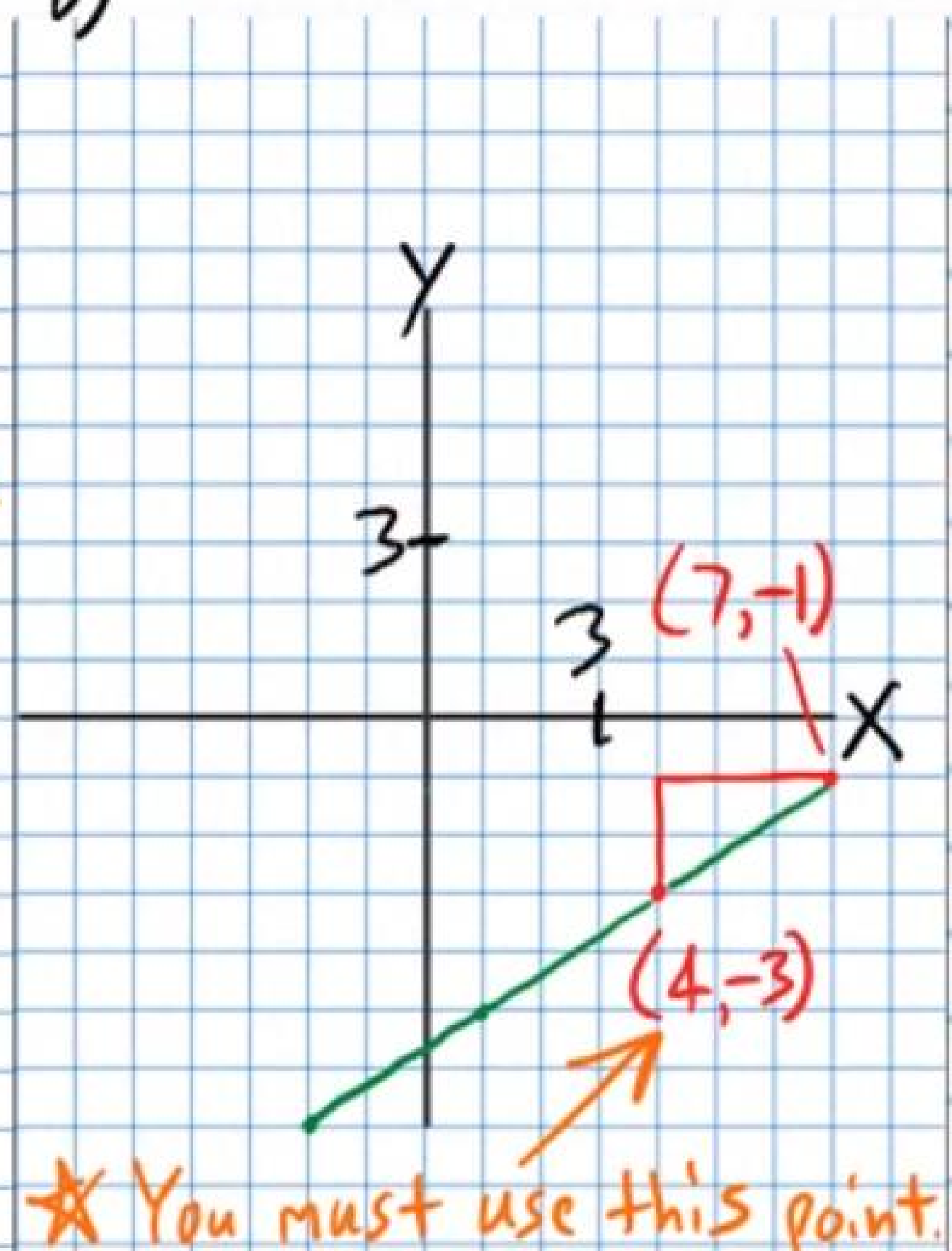
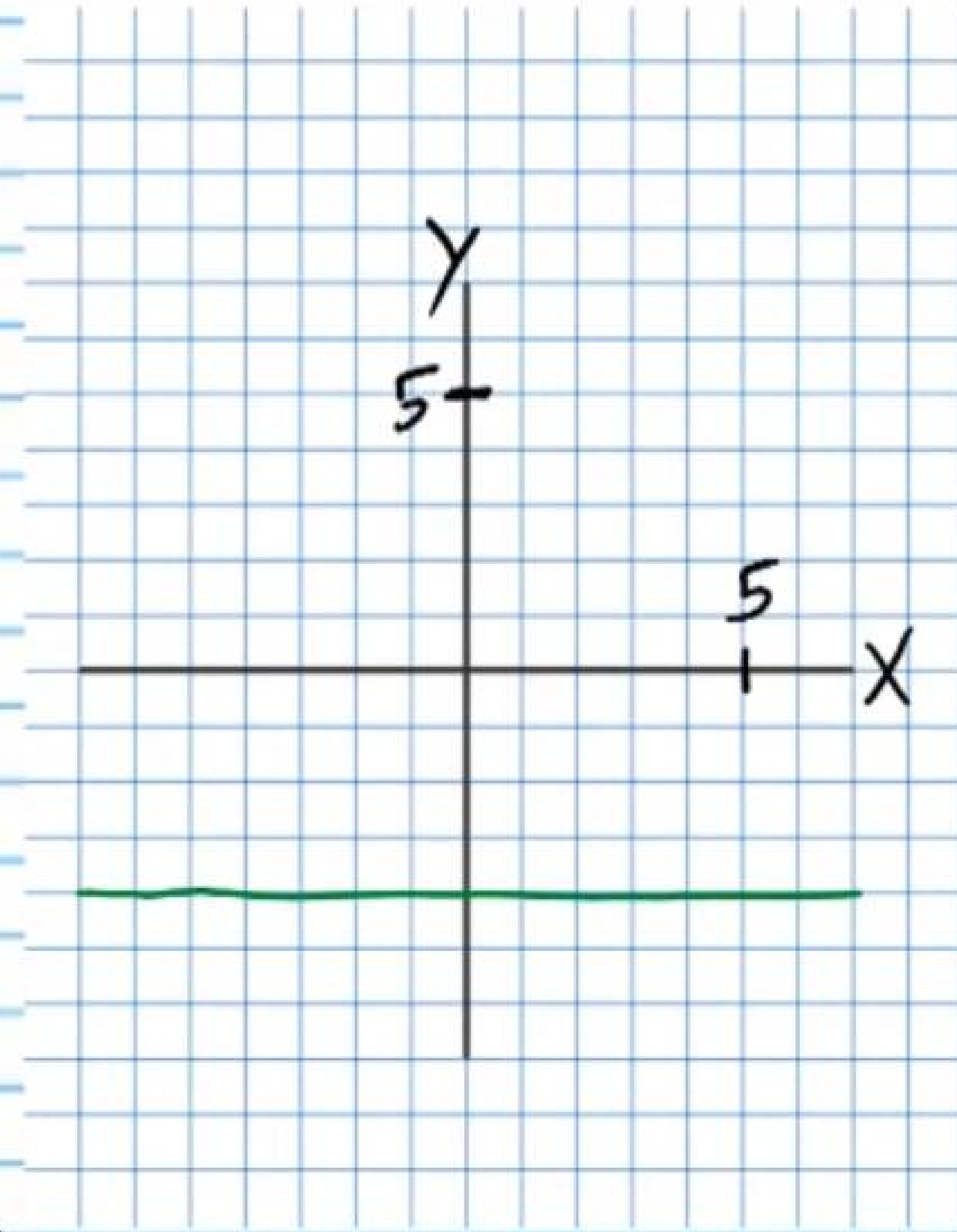


20) a)



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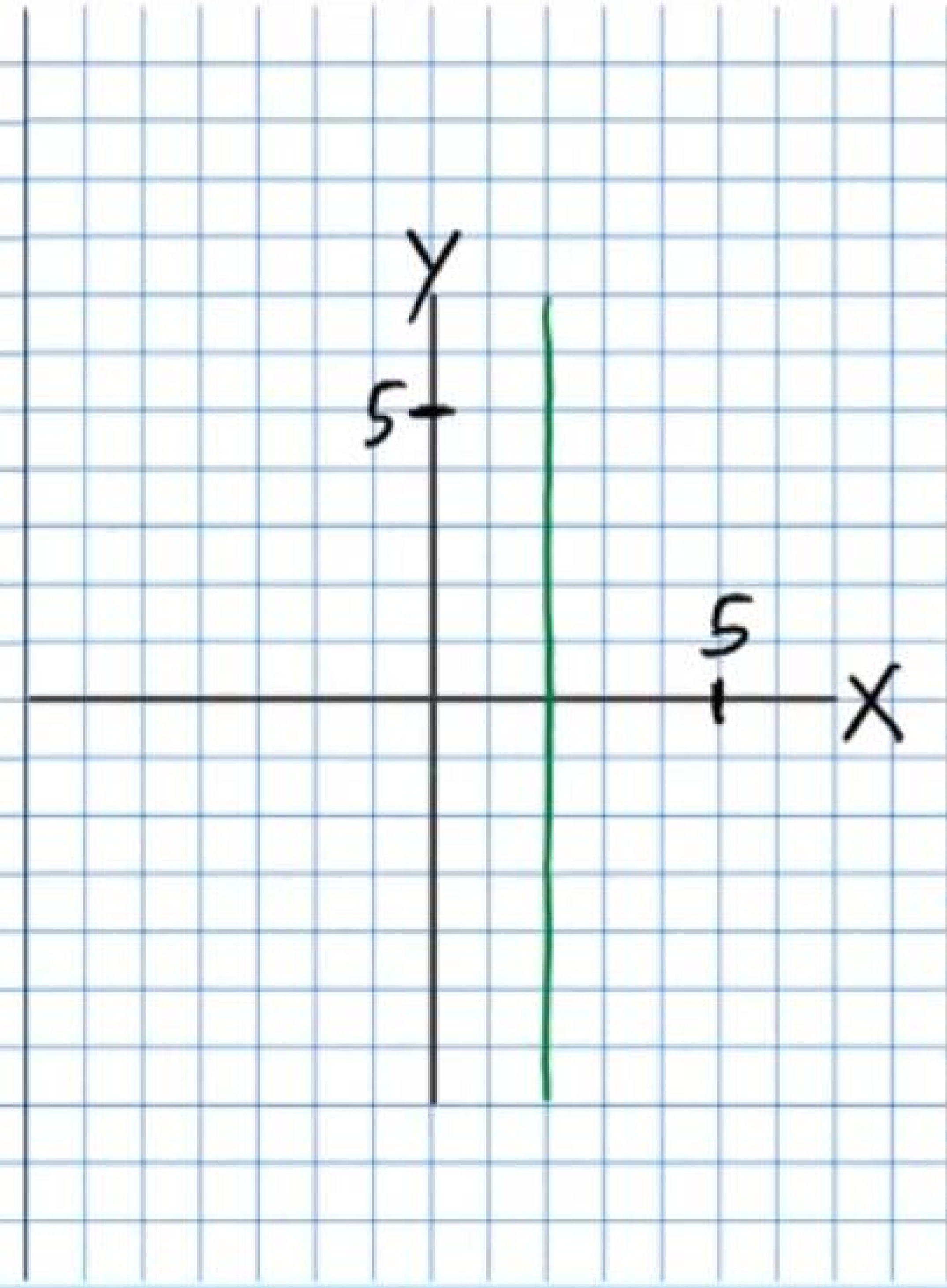

21) 9cm, 10cm, 11cm, 14cm

22) N = 9   Q = 14

23) 4.95 Liters of 90% fruit juice mixture
4.05 Liters of 50% fruit juice mixture

24) 13.5 oz of \$0.12/oz lemonade
16.5 oz of \$0.32/oz lemonade

25) 16cm X 33cm